

Qassim university

College of Computer

Department of computer science



CS471 – Web Technologies

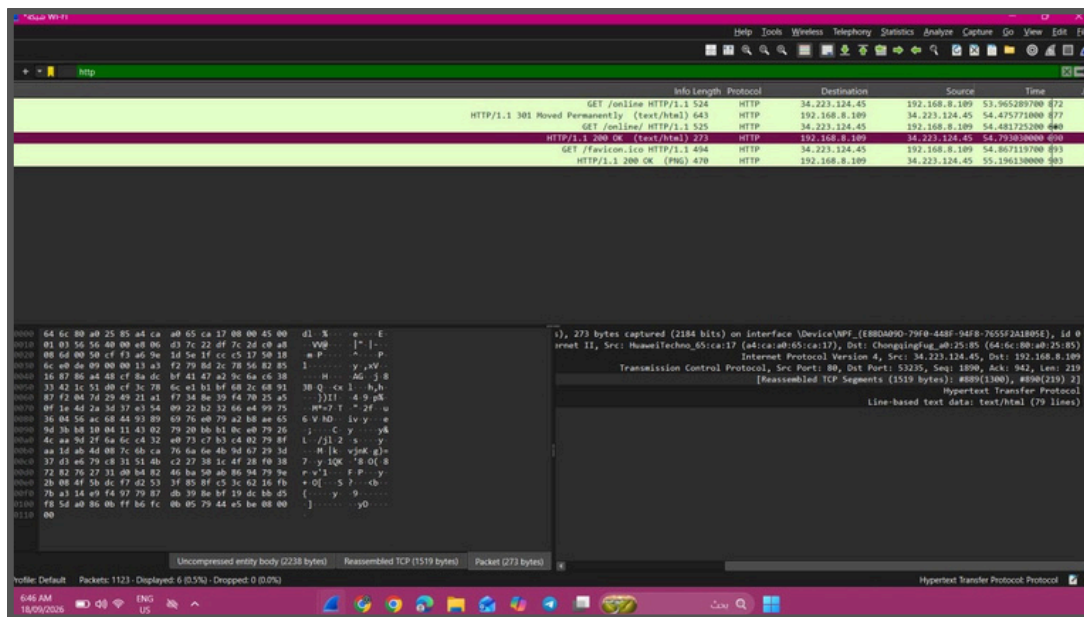
Lab 1

**CS471 – Web Technologies
(Laboratory)**

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Part 1 Capturing HTTP Traffic

Task 1 & Task 2: Filter and Analyze HTTP Packets

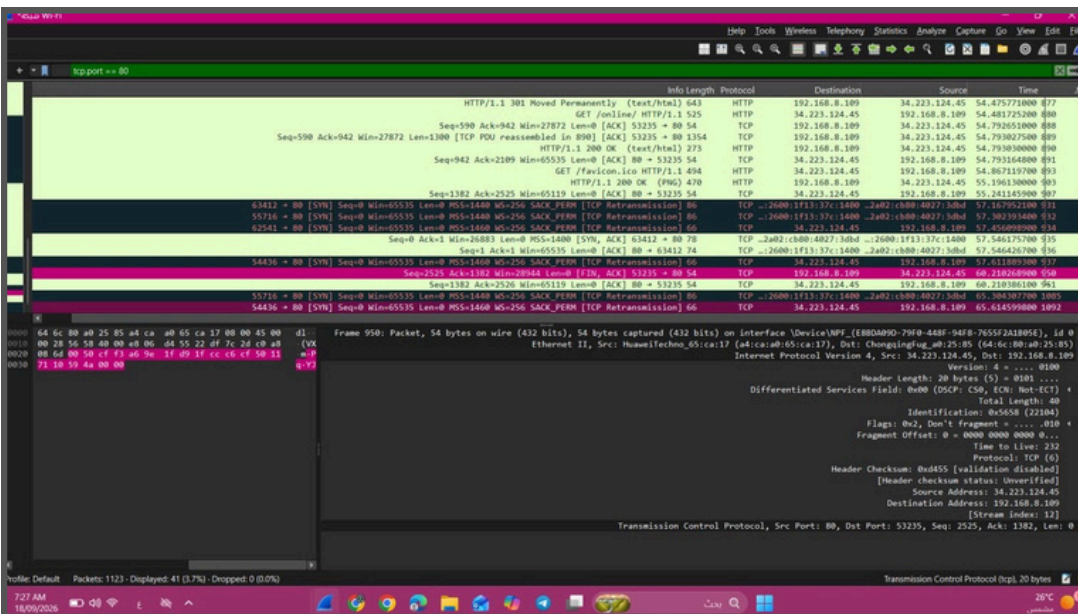


Captured unencrypted HTTP traffic from neverssl.com.

And Applied the http filter to inspect requests and responses. Observed HTTP request methods like GET /online and response status codes such as 200 OK and 301 Moved Permanently. Standard HTTP traffic transfers plain-text data without encryption.

Part 2: Analyzing TCP/IP Traffic

Task 1: Filter TCP Packets



Filtered packets using tcp.port == 80 and used the Follow -> TCP Stream feature to view the entire stream conversation between the client (192.168.8.109) and the server (34.223.124.45).

Task 2: Analyze TCP Handshake, Data Transfer, and Termination

63412 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 86	TCP → 2600:1f13:37c:1400 → 2a02:cb80:4027:3dbd 57.167952100 931
55716 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 86	TCP → 2600:1f13:37c:1400 → 2a02:cb80:4027:3dbd 57.302393400 932
62541 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 66	TCP 34.223.124.45 192.168.8.109 57.456098900 934
Seq=0 Ack=1 Win=26883 Len=0 MSS=1400 [SYN, ACK] 63412 → 80 78	TCP → 2a02:cb80:4027:3dbd → 2600:1f13:37c:1400 57.546175700 935
Seq=1 Ack=1 Win=65535 Len=0 [ACK] 80 → 63412 74	TCP → 2600:1f13:37c:1400 → 2a02:cb80:4027:3dbd 57.546426700 936
54436 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 66	TCP 34.223.124.45 192.168.8.109 57.611889300 937

Three-Way Handshake / Connection established using SYN (client request), SYN-ACK (server response), and ACK (client acknowledgment) packets.

Seq=1 Ack=471 Win=26880 Len=0 [ACK] 53235 → 80 54	TCP 192.168.8.109 34.223.124.45 54.271072700 876
HTTP/1.1 301 Moved Permanently (text/html) 643	HTTP 192.168.8.109 34.223.124.45 54.475771000 877
GET /online/ HTTP/1.1 525	HTTP 34.223.124.45 192.168.8.109 54.481725200 880
Seq=590 Ack=942 Win=27872 Len=0 [ACK] 53235 → 80 54	TCP 192.168.8.109 34.223.124.45 54.792651000 888
Seq=590 Ack=942 Win=27872 Len=1300 [TCP PDU reassembled in 890] [ACK] 53235 → 80 1354	TCP 192.168.8.109 34.223.124.45 54.793027500 889
HTTP/1.1 200 OK (text/html) 273	HTTP 192.168.8.109 34.223.124.45 54.793030000 890
Seq=942 Ack=2109 Win=65535 Len=0 [ACK] 80 → 53235 54	TCP 34.223.124.45 192.168.8.109 54.793164800 891
GET /favicon.ico HTTP/1.1 494	HTTP 34.223.124.45 192.168.8.109 54.867119700 893

Data Transfer / Observed HTTP requests/responses exchanged using Sequence (Seq) and Acknowledgment (Ack) numbers to maintain packet order and delivery integrity.

54436 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 66	TCP 34.223.124.45 192.168.8.109 57.611889300 937
Seq=2525 Ack=1382 Win=28944 Len=0 [FIN, ACK] 53235 → 80 54	TCP 192.168.8.109 34.223.124.45 60.210268900 950
Seq=1382 Ack=2526 Win=65119 Len=0 [ACK] 80 → 53235 54	TCP 34.223.124.45 192.168.8.109 60.210386100 951
55716 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 86	TCP → 2600:1f13:37c:1400 → 2a02:cb80:4027:3dbd 65.304307700 1085
54436 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [TCP Retransmission] 66	TCP 34.223.124.45 192.168.8.109 65.614599800 1092

TCP Termination / Identified connection closure handled gracefully via FIN, ACK and final ACK packets.

Part 3: Capturing and Analyzing UDP Traffic

Task 1 & Task 2: Filter and Analyze UDP Packets

Standard query 0x8b64 AAAA chrome.google.com 77	DNS 192.168.8.1 192.168.8.109 3.769005000 16
Standard query 0x91e1 A chrome.google.com 77	DNS 192.168.8.1 192.168.8.109 3.769473300 17
a7d6 HTTPS chrome.google.com CNAME www3.l.google.com SOA ns1.google.com 148	DNS 192.168.8.109 192.168.8.1 3.888970400 18
chrome.google.com CNAME www3.l.google.com AAAA 2a02:1450:4006:810::200e 126	DNS 192.168.8.109 192.168.8.1 3.888973200 19
nse 0x91e1 A chrome.google.com CNAME www3.l.google.com A 142.251.39.174 114	DNS 192.168.8.109 192.168.8.1 3.888974400 20
Len=401 443 → 53226 443	UDP 216.239.38.120 192.168.8.109 4.542137100 44
Len=200 443 → 53226 242	UDP 216.239.38.120 192.168.8.109 4.542627400 42
Len=29 53226 → 443 71	UDP 192.168.8.109 216.239.38.120 4.708159600 44
Len=245 53226 → 443 287	UDP 192.168.8.109 216.239.38.120 4.708670200 45
Len=1246 53226 → 443 1288	UDP 192.168.8.109 216.239.38.120 4.708678700 46
Len=221 53226 → 443 263	UDP 192.168.8.109 216.239.38.120 4.708680300 47
Len=26 53226 → 443 68	UDP 192.168.8.109 216.239.38.120 4.708838900 48
Len=26 53226 → 443 68	UDP 192.168.8.109 216.239.38.120 4.708839800 49
Len=33 443 → 53226 75	UDP 216.239.38.120 192.168.8.109 4.709009900 50
Len=33 443 → 53226 75	UDP 216.239.38.120 192.168.8.109 4.709239500 51
Len=33 443 → 53226 75	UDP 216.239.38.120 192.168.8.109 4.709359500 52
Len=33 443 → 53226 75	UDP 216.239.38.120 192.168.8.109 4.709477200 53
Len=274 53226 → 443 316	UDP 192.168.8.109 216.239.38.120 4.710186900 54

Frame 45: Packet, 287 bytes on wire (2296 bits), 287 bytes captured (2296 bits) on interface \Device\NPF_{E8BDA09D-79F0-448F-94F8-7655F2A1B05E}, id 0

Ethernet II, Src: HuaweiTechno_65:ca:17 (a4:ca:a0:65:ca:17), Dst: ChongqingFug_a0:25:85 (64:6c:80:a0:25:85)

Internet Protocol Version 4, Src: 216.239.38.120, Dst: 192.168.8.109

User Datagram Protocol, Src Port: 443, Dst Port: 53226

Source Port: 443

Destination Port: 53226

Length: 253

Checksum: 0x50ba [unverified]

[Checksum Status: Unverified]

[Stream index: 3]

[Stream Packet Number: 4]

[Timestamps]

UDP payload (245 bytes)

Analyzed Frame 45 showing a UDP response from the server (216.239.38.120:443) to the client (192.168.8.109:53226). The frame contains a total UDP length of 253 bytes (payload: 245 bytes) with a basic header structure lacking sequence tracking or connection flags.

Part 4: Comparing TCP and UDP

Task 1: Reliability and Connection Establishment

Criterion	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented, establishing connections via a 3-way handshake (SYN, SYN-ACK, ACK). It guarantees packet delivery through acknowledgments and retransmissions. UDP is connectionless and doe
Data Integrity and Ordering	TCP	TCP uses sequence numbers to ensure packets are reassembled in the correct order and performs error checking to retransmit lost or corrupted packets. UDP sends datagrams independently without sequence tracking or ordering

Task 2: Identify the Use Cases and Performance of TCP and UDP

Feature	TCP	UDP
Uses Cases	Web Browsing (HTTP/HTTPS), File Transfers (FTP), Email (SMTP/IMAP), Remote Access (SSH).	Video Streaming, Online Multiplayer Gaming, VoIP calls, DNS queries.
Performance	Slower Overhead: Higher latency due to connection handshake, 20-byte minimum header size, flow control, and congestion control algorithms.	Faster & Low Latency: Fast transmission with no connection setup and a minimal 8-byte header size.